



**A**  
**PROJECT REPORT**  
**ON**

“FocusLink – An AI Powered ADHD checklist app”

BY

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Submitted in Partial Fulfillment of

**Bachelor's of Data Science**

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***UNDER THE GUIDANCE OF***

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## **CERTIFICATE**

This is to certify that

**Bianca Antonia Khambatta**

has successfully completed their project on

**“FocusLink – An AI Powered ADHD Checklist App”**

in partial fulfillment of

Bachelor’s of Data Science

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# Student's Declaration

I undersigned, Bianca Khambatta, a student of Bachelor's of Data Science semester-VI, Department of Technology, Savitribai Phule Pune University, declare that the research project titled "FocusLink – An AI Powered ADHD Checklist App" is a result of my own work and my indebtedness to other work publications, references, if any have been duly acknowledged.

If I am found guilty of copying any other report or published information and showing as my original work, I understand that I shall be liable and punishable by Institute or University, which may include Failure in examination, repeat study and re-submission of the report or any other punishment that Institute or University may decide.

Bianca Antonia Khambatta

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Signature:

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# Abstract

FocusLink is a mobile application designed to support individuals with Attention Deficit Hyperactivity Disorder (ADHD) in building sustainable daily habits. Recognizing that most conventional habit trackers overlook the unique cognitive and emotional challenges faced by neurodivergent users, FocusLink integrates ADHD-aware design principles to create a calming, low-friction user experience.

The app was developed using React Native and Firebase, with OpenAI's GPT-3.5 model integrated for personalized suggestions and conversational support. Key features include a daily intention prompt ("The Big One"), mood-based habit generation using emoji inputs, a friendly AI chatbot coach, randomized encouragement notifications, and visual reward modals.

Each feature was deliberately chosen to address ADHD-specific barriers such as task initiation, emotional dysregulation, working memory deficits, and time blindness. By focusing on positive reinforcement and cognitive compassion rather than rigid productivity systems, FocusLink reimagines what a habit tracker can be — not a tool for discipline, but a digital ally for neurodivergent well-being.

This thesis details the design rationale, technical implementation, and impact of the app, highlighting how software can be both intelligent and empathetic when guided by inclusive design thinking.

# 1.Introduction – Addressing ADHD With FocusLink

Attention Deficit Hyperactivity Disorder (ADHD) is a widely prevalent neurodevelopmental condition that affects attention regulation, task initiation, and executive functioning. Individuals with ADHD often experience difficulty building consistent habits due to challenges with working memory, emotional regulation, and delayed reward systems. In today's fast-paced world, traditional productivity tools and generic habit-tracking apps fail to address the specific needs of neurodivergent users.

**FocusLink** is a mobile application designed to empower individuals with ADHD by providing a structured, emotionally supportive, and neurologically considerate environment for habit formation. Built using React Native and Firebase, FocusLink serves as an intelligent, low-friction daily companion that encourages consistency without overwhelming the user.

What makes FocusLink different from conventional apps is its deliberate use of ADHD-aware design elements — such as visual reward feedback, minimal decision fatigue through habit suggestions, and support for emotional check-ins. By integrating AI, motivational nudges, and a gentle interface, the app seeks to reduce the barriers that neurodivergent users face in forming sustainable routines.

This thesis documents the conception, design, and implementation of FocusLink. It explores how each feature — from “The Big One” prompt to AI-generated habit ideas — was chosen to address the psychological and behavioral dimensions of ADHD. The app was developed as a final-year project with a goal not only to showcase technical competence, but to solve a real-world problem through empathy-driven software.

The following chapters present a literature review of ADHD behavior and habit psychology, the rationale for each interface and logic decision, technical implementation using open-source tools, and user-centered evaluation. Through this, we aim to demonstrate how targeted UX combined with lightweight AI can produce a meaningful assistive technology for everyday mental well-being.

## **2. Literature Review – ADHD and Habit Formation**

Attention Deficit Hyperactivity Disorder (ADHD) affects approximately 5–7% of children worldwide, with symptoms persisting into adulthood in nearly two-thirds of cases. ADHD is characterized by difficulties in sustaining attention, managing impulsivity, regulating emotional responses, and initiating or completing tasks. These core challenges significantly interfere with an individual's ability to develop and maintain habits — especially those that require long-term consistency and delayed gratification.

According to Barkley's Executive Function Theory (1997), individuals with ADHD exhibit impairments in self-regulation, working memory, and goal-directed persistence. These deficits translate into poor habit-building capacity. For example, a neurotypical individual may gradually develop a habit through routine and reinforcement, while a person with ADHD may abandon the same habit due to forgetfulness, boredom, or emotional dysregulation. As a result, ADHD users often cycle through systems, apps, and planners without long-term adherence.

Recent research in behavioral science emphasizes that for ADHD-friendly habit systems to be effective, they must incorporate low-friction design, immediate reward feedback, minimal working memory load, and emotional engagement. Apps that bombard users with excessive



notifications, multi-step configuration processes, or unstructured dashboards often overwhelm rather than support ADHD users.

Furthermore, a 2020 study published in *Frontiers in Psychology* highlights the importance of personalization and emotional scaffolding in behavior change tools for neurodivergent populations. This research underscores the effectiveness of incorporating emotion-based check-ins and visual motivation systems — features central to the FocusLink app.

The role of dopamine in ADHD is also critical to habit science. Individuals with ADHD often experience dysregulated dopamine levels, which impacts motivation and reward processing. This necessitates a stronger emphasis on immediate positive reinforcement — something that FocusLink addresses through its visual rewards, emoji-based emotional prompts, and random encouraging notifications.

Several existing habit apps such as Habitica, Streaks, and Productive offer generic checklists with gamification. While some of these apps attempt to address motivation, few directly consider executive function deficits or emotional variability inherent in ADHD. FocusLink bridges this gap by blending clinical insights with accessible technology. It does not aim to “discipline” the user but instead gently supports and uplifts them throughout their day.

This chapter establishes the theoretical framework upon which FocusLink is built. By anchoring its design in psychological research and ADHD-specific behavior science, the app positions itself not just as a productivity tool — but as an ally in self-management for a deeply underserved population.

## 3. The Design Process – Why Each Feature Matters

The success of FocusLink is rooted not in complexity, but in intentional simplicity. Every feature in the app was chosen to serve a specific cognitive or emotional need of users with ADHD. This chapter outlines the rationale behind each major component of the user experience, linking it directly to common ADHD challenges such as low executive functioning, emotional dysregulation, working memory overload, and time blindness

### 3.1 The Big One Prompt

One of the most defining features of FocusLink is “**The Big One**”, a simple text input that appears at the top of the home screen each day. It asks the user: “*What’s the one big thing you want to do today?*”

This prompt targets one of the most significant ADHD pain points — difficulty with **prioritization and goal-setting**. By focusing attention on a single intention, the app reduces the paralysis caused by too many choices. The text input is visually separated from the checklist to give it emotional weight, and the user’s answer is persistently displayed as a subtle cue throughout the day. Unlike multi-step planning tools, “The Big One” is cognitively light and emotionally grounding.

## 3.2 Mood-Based Emoji Suggestion Bar

The app features an emoji bar where users can tap an emoji to reflect their mood (e.g., 😞, 😐, 😊, 😄). Upon tapping, an AI assistant generates a simple, customized habit based on their current state of mind.

This design serves multiple purposes:

- Provides **emotional scaffolding**, allowing the app to “meet the user where they are”
- Offers **context-aware suggestions**, rather than a rigid routine
- Supports **dopaminergic stimulation** through novelty and agency

Unlike traditional to-do lists that impose structure, this feature empowers users to create structure *in response to their feelings*, a critical accommodation for ADHD users who experience frequent emotional shifts.

## 3.3 AI Chatbot (The Coach)

The integrated chatbot allows users to ask for suggestions, ideas, or encouragement. It leverages OpenAI's GPT model to simulate a friendly, ADHD-aware coach who responds in short, affirming tones.

This coach is not just a productivity gimmick; it's a **social-emotional buffer**. Many ADHD users face emotional rejection sensitivity and benefit from gentle, validating interactions. The chatbot is a safe, judgment-free companion available any time the user feels stuck or overwhelmed.

### 3.4 Visual Rewards (Completion Modal)

When a user completes 3 habits in a day, a celebratory modal appears with a short message like:

“🎉 You completed 3 habits! Great job!”

While minimal, this form of **positive reinforcement** activates dopamine pathways that are often under-responsive in ADHD brains. It provides a clear, satisfying conclusion to behavior, reinforcing repetition without the pressure of streaks or perfectionism.

### 3.5 Random Encouragement Notifications

Every hour, while the app is running, a gentle nudge is sent to the user. Examples include:

- “You’re doing great — keep going 🧘”
- “Hydration check 💧”
- “Have you stretched today?”

These nudges **interrupt time blindness** and **simulate external structure**, which many ADHD individuals rely on. Unlike strict reminders, these notifications feel like support — not commands — and are randomized to avoid habituation.

### 3.6 Checklist Habit System

The core of FocusLink is a habit checklist that resets daily. Each item can be checked off with a tap and deleted with a long press. The list supports habit linking, allowing the user to relate one habit to another.

Unlike apps that pressure users with streaks or gamification, FocusLink's checklist is **low-stakes, frictionless, and calm**. It embraces imperfection and offers a nonjudgmental space for building routines — one habit at a time.

Every element of FocusLink was designed through the lens of cognitive compassion. It is not a tool to discipline neurodivergence, but to support and celebrate it. The features combine clinical insight, behavior design, and practical coding to deliver a uniquely gentle and impactful daily support system for ADHD users.

## 4. Implementation – How the App Was Built

FocusLink was developed using the **React Native** framework for cross-platform mobile development, enabling fast deployment to both Android and iOS environments. The app also leverages **Firebase** as its backend for real-time data storage, and **OpenAI's GPT-3.5-turbo API** for intelligent habit suggestions and chatbot communication. The implementation process was deliberately modular, prioritizing scalability, responsiveness, and minimal friction for the user.

## 4.1 Technology Stack

FocusLink was built using a carefully selected set of tools, each chosen to maximize accessibility, efficiency, and ADHD support:

- **React Native** was used for mobile app development, enabling simultaneous deployment to both Android and iOS platforms through a unified codebase.
- **Firebase Firestore** served as the cloud-based backend, allowing real-time storage and retrieval of user habits, habit history, and preferences.
- **Expo Notifications** handled local notification scheduling for the hourly encouragement system, designed to nudge users gently without overwhelm.
- **OpenAI's GPT-3.5-turbo** was integrated to provide real-time, personalized suggestions and chat responses, making the app feel intelligent and responsive to emotional states.
- **React Native StyleSheet** was used for clean, accessible interface styling, ensuring visual clarity and minimizing distractions.

## 4.2 Core Components

### HomeScreen.js

The primary screen of the app includes:

- Daily habit checklist
- “The Big One” priority input

- Mood-based emoji prompt bar
- AI Chat button
- Visual reward modal

Each of these components is built with React Native hooks (`useState`, `useEffect`) to support reactivity and state preservation across the session. Data for habits is fetched from and written to Firebase in real-time using `getDocs`, `addDoc`, `updateDoc`, and `deleteDoc`.

### **useEncouragements.js**

This custom hook uses `setInterval` and Expo's local notifications to send an encouraging message every hour while the app is running. It pulls randomly from a list of affirming prompts and triggers a local notification via `Notifications.scheduleNotificationAsync()`.

This file is imported into `HomeScreen.js` and invoked on render, requiring only one line of code for full functionality.

### **ChatScreen.js**

The AI chatbot screen sends user queries to the GPT-3.5-turbo API via Axios. The returned message is formatted and displayed inside a scrollable chat window. This screen provides emotional support and problem-solving suggestions in real-time.

## **4.3 API Integration**

The app uses OpenAI's GPT endpoint as follows:

This is used both for mood-based habit suggestions and the AI coach chat.

```
const res = await axios.post(
  'https://api.openai.com/v1/chat/completions',
  {
    model: 'gpt-3.5-turbo',
    messages: [
      {
        role: 'system',
        content: 'You are a friendly, creative ADHD-aware habit coach. Keep responses under 12 words.',
      },
    ],
  },
)
```

## 4.4 UI Design Philosophy

The interface uses a soft, readable color palette, emojis, and rounded containers to create a visually calming environment. Typography (via Inter and Open Sans fonts) is chosen for clarity and approachability. Interactive elements such as the emoji row, checklist, and input fields follow a “low effort” model — the user never has to guess or work to use the app.

Reward elements (modal pops, checkmarks) provide emotional closure, while AI interactions are short, friendly, and validating. The design avoids sensory overload and maintains a **single-column flow**, ideal for mobile ADHD use.

## 4.5 Performance Optimization

- The app minimizes re-renders by structuring components to only update when state changes.
- Lists are optimized via FlatList, which lazily loads habit entries.
- Firebase access is asynchronous and structured with try-catch blocks to prevent UI blocking.



FocusLink's technical build showcases that **simplicity, empathy, and intelligence** can be combined in a lightweight mobile app. The technology was chosen not for its complexity, but for its ability to meet the emotional and cognitive needs of a specific user base.

## **5. Evaluation – Strengths, Limitations, and Future Improvements**

FocusLink's evaluation centered on its ability to meet the emotional, functional, and cognitive needs of individuals with ADHD. Initial user feedback highlighted the app's simplicity, visual clarity, and emotional resonance as major strengths. Users especially appreciated features like 'The Big One' for daily clarity, the randomized encouragements, and the non-intrusive reward system.

One strength of the application is its low cognitive load. The use of emojis, one-tap actions, and minimalistic layout allows users to navigate and complete tasks without friction. Additionally, the mood-based AI habit suggestion model creates a sense of personalization and responsiveness.

However, the app is not without limitations. Currently, the habit data and 'Big One' entry do not persist across multiple days or devices, which may limit long-term usage. Furthermore, while the app includes a chatbot, it lacks emotional memory or advanced personalization.

Planned improvements include persistent storage of 'The Big One', enhanced habit analytics, and push notification scheduling for greater flexibility. Integrating journaling and community features may also enhance user support and long-term engagement.

## **6. Impact – How FocusLink Helps Real ADHD Users**

FocusLink addresses core ADHD challenges such as task initiation, emotional regulation, and executive functioning by translating clinical principles into practical features. The app does not attempt to 'fix' ADHD but rather to design around it, respecting the unique needs of neurodivergent minds.

By enabling users to set a single intention each day through 'The Big One', FocusLink reduces cognitive overwhelm and fosters intentional action. The emoji-based habit prompts help users honor their feelings rather than push through them blindly. Encouraging nudges interrupt time blindness, while visual rewards support dopamine activation in a gentle and non-gamified way.

Ultimately, the app empowers ADHD users to feel seen, supported, and capable — not by enforcing discipline, but by offering a friendly system they can rely on. Feedback from users consistently reflects appreciation for how the app validates their experience and supports meaningful, sustainable routines.

## 7. Conclusion

FocusLink was developed with a clear purpose: to provide a meaningful, ADHD-aware solution for habit-building that embraces compassion, emotional intelligence, and practical functionality. In contrast to conventional productivity tools that demand conformity, FocusLink gently adapts to the neurodiverse mind, helping users thrive on their own terms.

This thesis documented the app's development journey, from psychological foundations to technical implementation. Each feature — from 'The Big One' to the AI-driven mood-based habit suggestions — was rooted in user-centered design that respects the lived experience of ADHD. The result is a mobile application that not only works, but resonates deeply with those it was created for.

Future iterations of FocusLink may include cloud syncing, journaling, advanced analytics, and community connection — but its core value will remain unchanged: to make habit-building emotionally safe and intuitively supportive for individuals with ADHD. The project demonstrates that when technology is guided by empathy and purpose, it becomes far more than a tool — it becomes a companion for personal growth.